

**NED UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**FE SEC A & B (BACHELOR OF SOFTWARE ENGINEERING)**  
**SPRING SEMESTER MIDTERM EXAMINATIONS 2024**

**TIME: 1.5 hours**

**DATED: 20/05/2024**  
**MAX MARKS: 20**

**ACADEMIC READING AND WRITING (EA-115)**

**Note: Attempt all questions in your answer sheets.**

**Q.1. Read the passage below and answer the questions under it. [CLO 1, 8 marks]**

**THE KOREAN WAVE**

The Asia Society Korea Center's Monthly Luncheon returned on Tuesday with a fascinating insight into Korean pop music from the President of DFSB Kollektive, Bernie Cho. Mr. Cho specializes in providing digital media, marketing, and distribution solutions to over 350 Korean Pop music artists and was kind enough to speak at the Korea Center on "A Case Study on K-Pop: The Hip Hype Reality of the Korean Wave". Mr. Cho first came to Korea in 1993 with the goal of pursuing his graduate studies at Yonsei University, however through a chance meeting at a Seoul nightclub ended up working on the launch of Korea's first music channel and his career in the music industry began. His talk focused on how the successes of the Korean music industry have stemmed from the frustrations those within the business have faced and how through diversification and glocalization they were able to flip these challenges and turn them into opportunities.

Mr. Cho explained how in 1993 the Korean music industry was very different from the one we see today; 70% of music in the country was international pop music with Korean trot and country music making up the other 30%. Fast forward to 2002 and K-Pop, as it was now known, accounted for 70% of the market, and Korean artist BoA was dominating the charts both in Korea and Japan. Today, Korea is ranked as the 8th largest music market in the world, was the first country in the world where digital sales outsold physical records, and is the 4th largest net music exporter behind the U.S., U.K., and Sweden. Mr. Cho talked about how these accolades came as a result of hard work and sweat born out of frustrations that the Korean music industry constantly faced.

Korea is the most wired nation in the world where cell phone penetration stands at 110% and the country is experiencing the digital age. Yet the digital music stores and services are dominated by Korean providers where overseas services such as iTunes and Spotify are prohibited. Mr. Cho explained how the problem lies in the price that these Korean portals charge for downloads; at 9 cents a song, which is 16 times cheaper than in the U.S and 32 times cheaper than in Japan, a lob-sided market has formed where a song made in Korea can make 23% more profit when sold overseas. Because of these market conditions, the Korean music industry has had to diversify and look at other opportunities to make money.

One opportunity lies in the overseas market where Korea has seen over 60% growth in exports over the past 5 years. Mr. Cho gave the example of Psy and his worldwide hit Gangnam Style as a benefit of tapping into the overseas marketplace. In Korea, the song had 3.6 million digital downloads and over 40 million streams, yet earned Psy only \$61,000 compared to the \$ 2.7 million he earned from roughly the same number of downloads and streams outside of Korea. S.M Entertainment and YG Entertainment, two of Korea's biggest music companies only produce 20-30% of revenue from music sales in Korea; more than 50% comes from overseas.

So why has Korea been so successful in the overseas market? Mr. Cho puts this down to "glocalization". The early 2000s saw a shift from Korean-made music for domestic consumption to Korean-made music marketed for international consumption. Today we even see music made by Koreans but written, recorded, and performed in countries such as the U.S and Sweden. Three factors have helped with this "glocalization"; the bilingual music being produced, multicultural K-pop groups with international members, and a boom in K-pop fan sites that dominate online portals.

Through the domestic challenges that those in the industry faced, Korean entrepreneurs were forced to explore other avenues to generate profit, of which the overseas market played a huge role. Mr. Cho summarized his talk by saying that without the influence of these Korean music entrepreneurs, the "Korean Wave" could well have turned into a "Korean Tsunami" and we would not have the booming industry we have today.

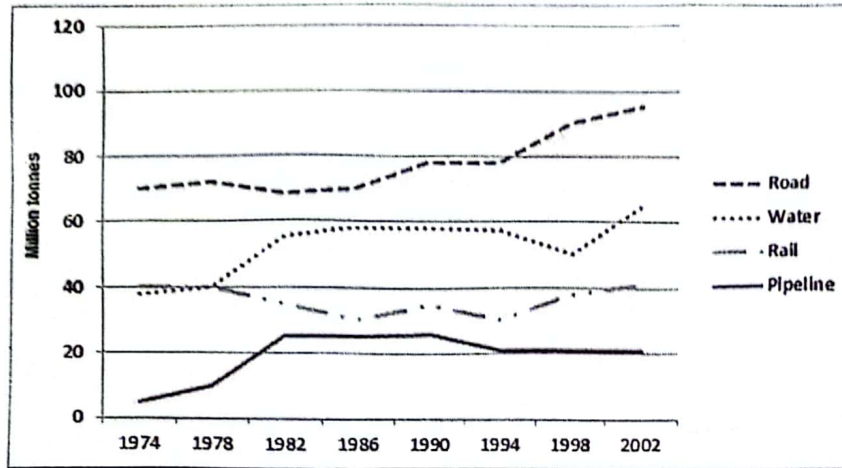
- 1. What was the state of the Korean music industry in 1993 and how has it changed over the years?**

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2. What are the three factors that have helped with the "glocalization" of Korean music?
3. Which Korean music artists have contributed to the Korean economy and how?
4. Describe another movement like the Korean Wave in any region that has impacted the social, cultural, political, or economic condition of the region.

**Q.3. Discuss the weight of the goods transported in the UK across the years stated along the x-axis, in the visual below. [CLO 1, 4 marks]**

**Goods transported in the UK (1974-2002)**



**Q.3. Brainstorm ideas on any 1 of the topics provided below using any one *relevant* pre-writing strategy. [CLO 2, 2 marks]**

- Reading fiction books can enhance our view of the world. Do you agree or disagree?
- Lower grades should be punished. Do you agree or disagree?
- Homework tasks help students retain knowledge better. Do you agree or disagree?

**Q.4. Write a 200-word (approx.) essay using the ideas you have brainstormed in Q.3. [CLO 2, 4 marks]**

**Q.5. Highlight the thesis and forecasting statements and the topic sentences of each of the body paragraphs. [CLO 2, 2 marks]**

Best of Luck!

Q:1 → 7.75/8

Q:2 → 3/4

Q:3 → 2/2

Q:4 → 4/4

Q:5 → 2/2

Total →  $\frac{18.75}{20}$

(19)



## ANSWER NO. 1

Yusuf Arif

1. In 1993, the state of Korean music industry was quite different from today's industry. The percentile allocation was that 70% constitutes of international pop music with Korean hit while 30% of country music makes up for rest of it. Moving to year 2002, K-Pop comprises of 70% of market, and Korean artist BoA was gaining superiority in both, Korea and Japan. Currently, it's 8<sup>th</sup> largest music market in the world. All these changes are a result of sheer hardwork and the fruit born out of frustrations and chaos that the Korean music industry constantly faced. + 1.75

x-----x

2. Factors that have assisted with the 'globalization' of Korean music are as follows: +2

1. Bilingual or multilingual music production
2. Collaboration of multi-cultural K-Pop groups with international members.
3. A boom in K-pop fan sites that gained dominance on online portals.

x-----x

3. Korean artist BoA have made contributions to Korean economy by his popularity in both, Korea and Japan. Also, Psy and his well-known 'Gangnam Style' benefitted the economy by tapping into overseas. This song had 3.6 million digital downloads and approx 40 million streams, but Psy earned \$61,000 as in comparison to the \$2.7 million he earned from the same number of streams and downloads, in countries other than Korea. +2

x-----x

4. Pakistan has also a weak structure of music production in history. Previously, the artists and songs were not well known in regions other than itself. With time, artists like Nadia Hussain, Atif Aslam, Rahat Fateh Ali Khan<sup>xx</sup> gained popularity. They even sang for different nations, for instance, India, our rival! But this wave improved our social, cultural, and economic condition, as lately songs like "Kanayacari" and "Pasoori" gained dominance overseas. Even, 'Pasoori' ranked first amongst other songs and was on trending for weeks. Coke studios and other studios are a benefit to Pakistan's economy, and the production of songs of different languages helps in cultural and social condition. Thus, 'Pakistan Wave' is of greater benefit for our nation. +2

Q: 1  $\rightarrow 7.75/8$

## ANSWER No. 2

Provided is a line graph, depicting the weight of goods transported in UK, across the years, 1974 till 2002. Roads bears the highest weight, with increase along the progression of years, while pipelines are the least weighed transported good. Its trend shows slight increase, then rapid increase, then constant line and lastly a decline, which remained constant from 1994 to 2002. Water and rail weighed 40 million tons till 1978, then water's weight increased till 1994 while rail had slight changes. There is a change in year 1998, in water line, after which it had constant increase in its weight.

All in all, roads are the most weighed transported good and pipelines are the inverse of it!

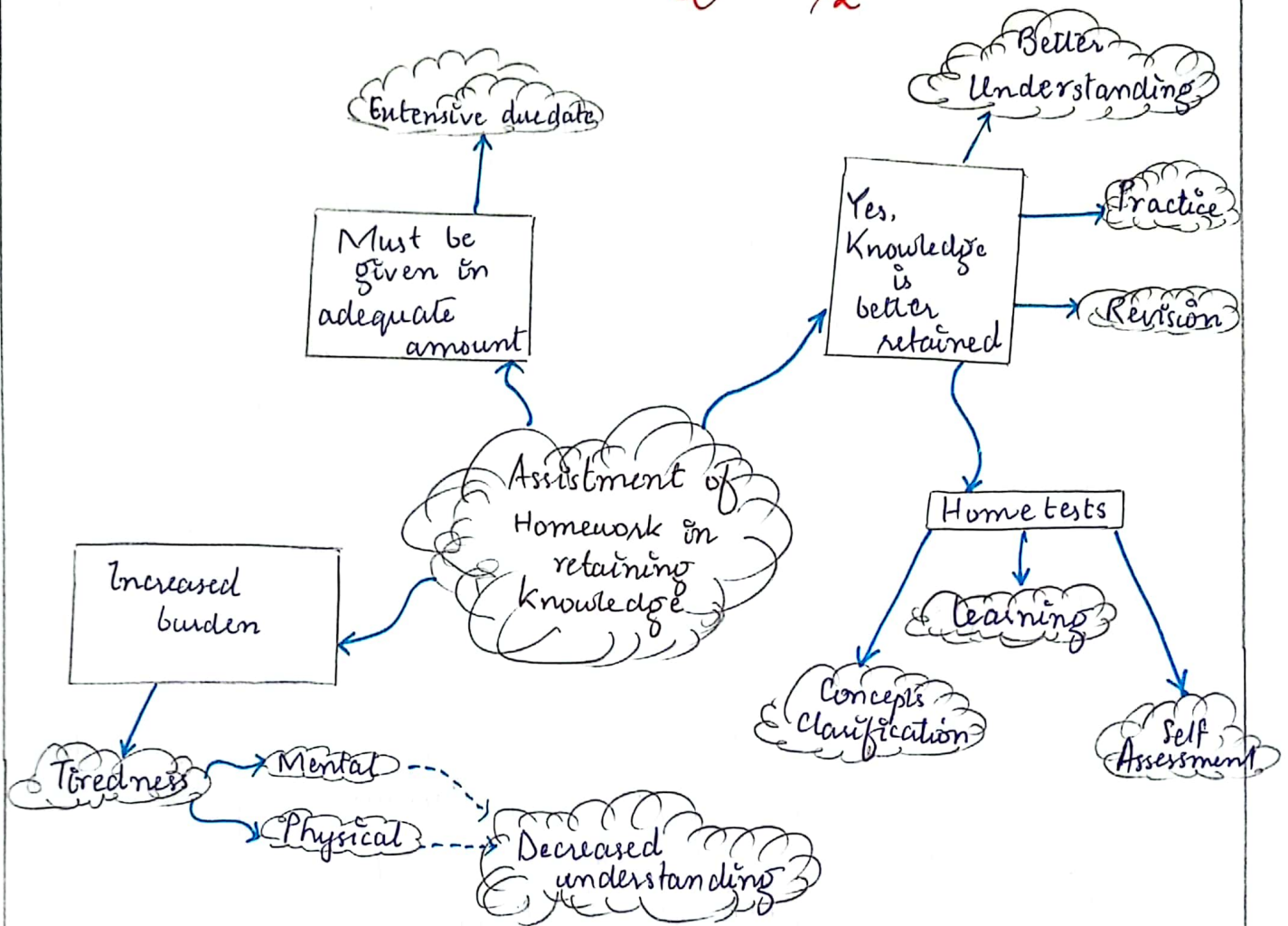
Q: 2  $\rightarrow 3/4$



## ANSWER No. 3

Topic: "Homework tasks help students retain knowledge better. Do you agree or disagree?"

Q:3 → 2/2



Brainstorming Technique: CLUSTERING

## ANSWER No. 4

### "ASSISTMENT OF HOMEWORK IN RETAINING KNOWLEDGE"

The assignment of homeworks given by teachers, to students has been issue of controversy that whether the students should be bombarded with tasks or not. It is a matter of discrepancy as some says, that knowledge is better retained, while other presents

a devil's advocate. The following essay highlights the positive and negative outcomes of homeworks.

Undoubtedly, homeworks are for student's own benefit as by completing these, they can have better understanding of the topic. Further, their concepts are polished, also, they are a means of practice. While completing homeworks of any domain, a student gets the broader and wider perspective of the topic under discussion, thus building his core concepts, which are retained for long period.

Additionally, homeworks containing quizzes or test, helps the student in self-assessment. It enhances his learning and information. But, the issue here is, that some tasks are graded, so students' focus shift towards marks rather than deep clarification. Homeworks should not be graded in my and many other's point of view which is somewhat correct.

On the contrast, home assignments can be a burden over student's weak shoulders. Increased tasks with rapid deliveries tires a student, both mentally and physically. As a result, his understanding capacity is lessened, and the topics he is reading or writing have an ephemeral effect on his mind.

In a nutshell, homeworks have the potential for better retainment. Increased understanding, enhanced learning and better interpretation and clarification of the topics. But only if, they have been assigned in adequate amount with proper due time and without grading. It purely depends upon the lecturer, that whether his assigned tasks can help in retaining knowledge better or the tasks can become burden resulting in poor understanding.

*Very well-written & thoughtful essay!*

Q:4 → 4/4

Q:5 → 2/2

ANSWER NO. 5

| INVENTORY: (Intro)     | (In intro para)             | (In 3 Body Para)      |
|------------------------|-----------------------------|-----------------------|
| ____: Thesis Statement | ____: Forecasting Statement | ____: Topic Sentences |

x The End x



- Q1. Giving reasons for your answer based on the type of system being developed, **propose** the most appropriate software process model that might be used as a basis for managing development of an interactive travel planning system that helps users plan journeys with the lowest environmental impact. [02] CLO2
- Q2. **Discuss** the problems you may encounter when you are required to gather requirements from multiple stakeholders, suggest why it is important to state user requirements and system requirements differently. [03] CLO1
- Q3. **Explain** Scrum, **compare** it with conventional plan based approach. The comparisons should be based on the effectiveness for planning the allocation of people to projects, estimating the cost of projects, maintaining team collaborations, and managing changes in project. [06] CLO4
- Q4. **Differentiate** generic software product development and custom software development also discuss four important attributes that all professional software should have. [04] CLO1
- Q5. List any 8 principles of agile methodologies and **explain** how these principles lead to the accelerated development and deployment of software. [05] CLO2

Suppo

Main Reason

- For location (real-time), interactive UI is required, which can be achieved by multiple development of prototypes.
- Prototype models have 'iterative nature', which can help in accommodating changes.
- As the software is step-by-step<sup>system</sup>, so it can easily be made by Prototyping, the customer may review the final product before the development of actual software, and can give feedback which proves beneficial for the developers.
- Users may suggest the most interactive, appealing and easiest system by giving suggestions in feedback.



ANSWER NO. 1

**SCENARIO:**

Interactive travel planning system that helps users plan journeys with lowest environmental impact.

**MODEL:** "PROTOTYPE - MODEL"**REASON:**

Supporting Reasons

- As this model, helps the user to better understand. As any man with having least knowledge of softwares may want to plan their journey. So, by using this model, they can easily visualize and understand the requirements by visuals, pictures, and designs.
- For location (real-time), interactive UI is required, which can be achieved by multiple development of prototypes.
- Prototype models have 'iterative nature', which can help in accomodating changes.
- As the software is step-by-step<sup>system</sup>, so it can easily be made by Prototyping, the customer may review the final product before the development of actual software, and can give feedback which proves beneficial for the developers.
- Users may suggest the most interactive, appealing and easiest system by giving suggestions in feedback.

Main Reason



## ANSWER No. 2

When we want to gather requirements from multiple stakeholders, the following issues are faced:

- It takes time to elicit requirements by multiple stakeholders.
- Facing difficulty in prioritization of requirements.
- Stakeholders may have contradicting requirements.
- Tackling with multiple customers or stakeholders is an art, not possessed by every developer.
- Becomes difficult to analyze the budget when there are many requirements by different stakeholders. Budget may go out! <sup>constraints</sup>
- More people's <sup>involvement</sup> leads to different point of views which results in conflicts.

It is important to state user requirements and system requirements differently as

- by having user requirement in a separate block can help developers in dealing with the requirements or features required by the system.
- System's requirements depends on user requirements (Dependency)
- One must state them differently while documenting so, customers or any person might not get confused and could easily understand the reason behind the use of that specific 'system' feature used by developer.

2/3

## SCRUM

Scrum is the most used agile methodology. It is suitable for complex projects. It is an effective way which demonstrates the collaboration of users and developers.

### When to Use?

- + Aggressive deadlines or limited timelines
- + Complex and large projects.
- + When rapid deliveries are required.
- +

### SCRUM ROLES

**PRODUCT OWNER:** The person who gives deadlines, requirements, etc. The one to whom the product belongs.

**SCRUM LEADER:** Cheerleaders of Scrum. They motivate and monitor that whether the team is following scrum practices etc.

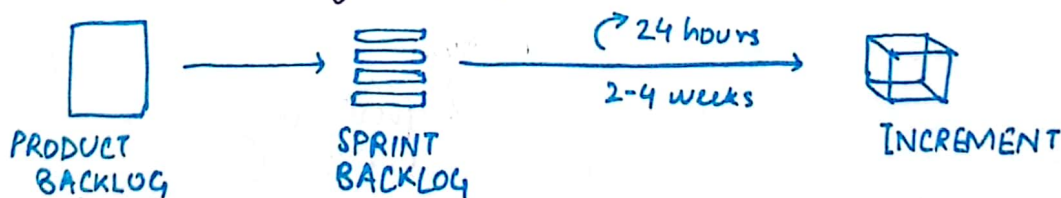
**DEVELOPMENT TEAM:** 4-9 members are in the development team. They all are self-driven individuals, who know programming, & designing both.

### SCRUM ARTIFACTS:

**PRODUCT BACKLOG:** Initial document given by product owner. It is basically the 'to-do-list'. It can be modified afterwards.

**SPRINT BACKLOG:** The tasks that are to be done firstly, are extracted from product backlog. Sprint backlog consist of tasks that are to be achieved in one sprint.

**INCREMENT:** Small but working model of the system. It is the final product at the end of each sprint.



### SCRUM PROCESS

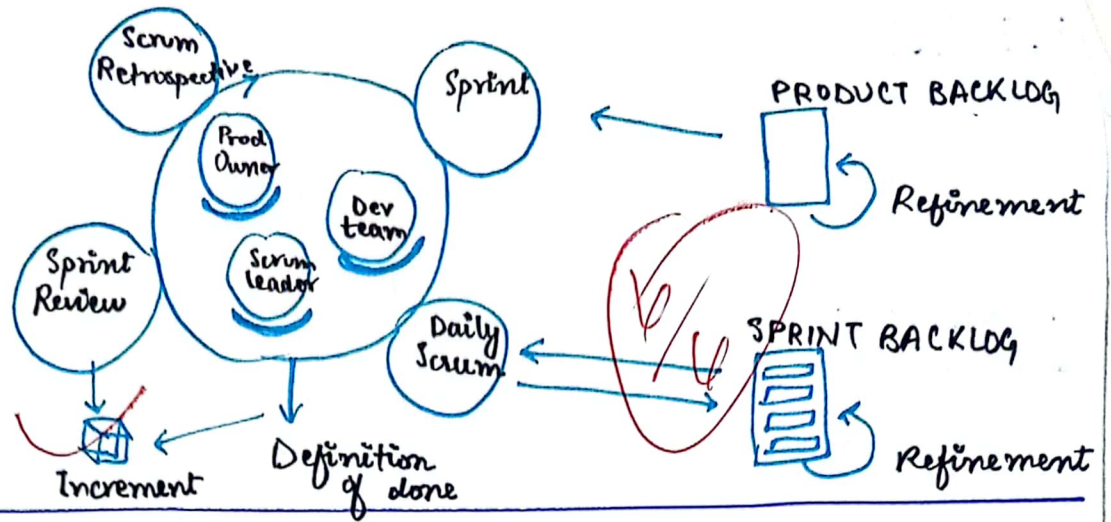
→ Firstly, sprint <sup>backlog</sup> is produced in which the goals, and objectives are discussed, and tasks are allocated.

→ Then, comes daily scrum which is a stand-up meeting (25 mins) in which the work of today and next day is discussed.

→ Thirdly, sprint review is done which is the meeting when a sprint is completed. Developers showcase their work and ask for improvements which will be made in next sprint.



## SCRUM PROCESS



→ lastly, a meeting hosted by scrum leader is held, & is known as scrum retrospective. It focuses on the problems & challenges faced by workers.

## COMPARISON WITH CONVENTIONAL PLAN BASED APPROACH:

1. In conventional plan-based approach, collaboration and communication between the customer and software developers is lessened, unlike scrum, where meetings are held between both.
2. The customer may not have better understanding in conventional approach while in scrum, the members keep the customer in loop.
3. Feedbacks are given in scrum in the phase of sprint review which enhances the effectiveness of the product, unlike normal plan based approach.
4. Changes can be accommodated easily in scrum, on the contrary, in conventional approach changes can even cost more than the actual budget of application.
5. Scrum can be expensive than conventional plan-based approach as it constitutes of a team, which delivers the software rapidly in less time. Conventional based approach usually costs less in comparison.

## GENERIC PROCESS DEVELOPMENT

This process involves the formation of generic product.

This process is less expensive.

Developers listen to themselves, & can make product according to their will.

The product is installed in less time

This process is short and consumes less time.

## CUSTOM SOFTWARE DEVELOPMENT

This process involves the formation of custom product.

It is more expensive.

In this process, developers are bounded by user's or customer's requirement & will and cannot make change to their will.

It takes more time in the installation as extensive or excessive features are also added, because of customer's choice.

This process is time-consuming & long.

3/4

## COMMON ATTRIBUTES:

- Maintainability: The software should be able to incorporate changes. It should be maintainable. Maintenance period is imp as developer will change for fixing bugs after it.
- Acceptance: Acceptance criteria is the user's perspective of the software. It should be at least 95%.
- Scalability: The requirements and software should be scalable.
- Security: The software should ensure that it is not accessed by unauthorized users. The information should be private and secured.

## ANSWER 5.

## AGILE METHODOLOGY'S PRINCIPLES:

There are 12 principles documented by the agile team. Some are as follows

1. Working software is a measure of progress.
2. Working software is delivered frequently.
3. Project team is revolved around motivated individuals.
4. Sustainable development is necessary to maintain constant pace.
5. Face to face communication is the best form of communication.



6. ~~Cooperation and collaboration~~ between developer and customer
7. Best architecture requirements and designs are achieved by self-organized teams.
8. Customer satisfaction is ~~necessary~~ and is achieved by rapid deliveries and valuable ~~software~~.

The principles lead to accelerated development and deployment of software as agile methodologies involve the collaboration by software engineers, customers, designers which helps in the completion of software product in less time, as requirements are gathered in reduced time, & as all members are motivated & passionate so there is fast delivery of the desired product.



SE - 080

**NED UNIVERSITY OF ENGINEERING & TECHNOLOGY**  
**DEPARTMENT OF SOFTWARE ENGINEERING BATCH(2023)**  
**FIRST YEAR FALL SEMESTER 2023 MID TERM TEST**

TIME: 30 MINS

DATED: 24-5-2024

**APPLIED PHYSICS PH-122****ATTEMPT ALL QUESTIONS****MAX. MARKS: 20**

|            |                                                                                                                                                                                                                                        |             |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Q1.</b> |                                                                                                                                                                                                                                        | <b>CLO1</b> |
| a)         | Discuss graphical techniques and advantages of converting nonlinear curve into linear curve?                                                                                                                                           | <b>3</b>    |
| b)         | What is scalar Triple product and vector Triple product show that:<br>$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$                                                    | <b>3</b>    |
| c)         | Define Gradient, Divergence, Curl, line integral?                                                                                                                                                                                      | <b>4</b>    |
| <b>Q.2</b> |                                                                                                                                                                                                                                        | <b>CLO2</b> |
| a)         | Show that $\nabla r^n = nr^{n-2} \vec{r}$ OR Find grad $\phi$ , if $\phi = \ln  r $                                                                                                                                                    | <b>3</b>    |
| b)         | If a particle is moving under the influence of force $\vec{F} = 3x^2 \hat{i} + (2xz - y) \hat{j} + z \hat{k}$<br>Find the work done $\int_C \vec{F} \cdot d\vec{r}$ where c is straight line from $P_1 (0, 0, 0)$ to $P_2 (2, 1, 3)$ . | <b>4</b>    |
| c)         | Find the percentage error in the area of an ellipse if 1% error is made in measuring the major and minor axis                                                                                                                          | <b>3</b>    |



ANSWER NO. 1

(a)

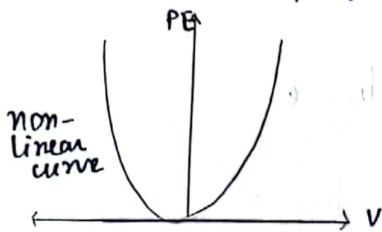
$$C_{101} = 10$$

$$C_{102} = 10$$

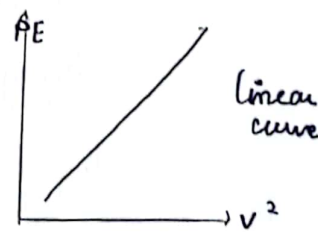
(20) f.h.

GRAPHICAL TECHNIQUES OF CONVERTING NON-LINEAR TO LINEAR :1. METHOD 1: PLOTTING QUANTITIES WITH POWER

Plotting quantities with power. For example, the graph between potential energy and  $v$  is non-linear, but if we draw with  $v^2$ , then it will be linear.



$$PE = \frac{1}{2}mv^2$$

2. METHOD 2: EXPONENTIAL FUNCTIONS

$$y = ae^{bx}$$

Taking log both sides,

$$\log y = \log ae^{bx}$$

$$\log y = \log a + \log e^{bx}$$

$$\log y = \log a + bx \log e$$

$$\therefore \log e = 1$$

$$\log y = \log a + bx$$

$$z = mx + c$$

where  $y = \log y$ ,  $m = b$ ,  $c = \log a$

$$b = \text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\log y_2 - \log y_1}{x_2 - x_1}$$

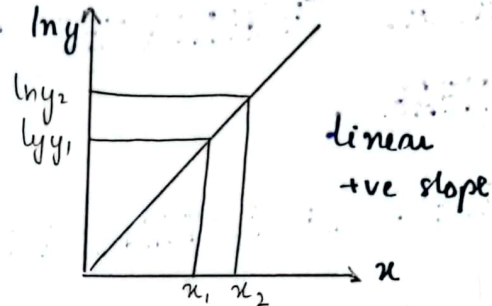
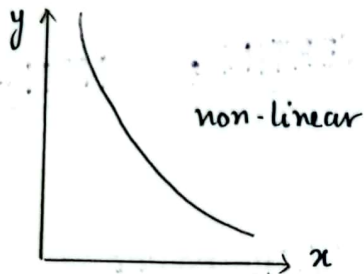
$$c = \log a$$

Take exp B.S

$$e^c = e^{\log a}$$

$$a = e^c$$

or  $a = e^c$



### METHOD 3: POLYNOMIAL FUNCTIONS

$$y = a \cdot x^b$$

Take log BS,  $\log y = \log a \cdot x^b$

$$\log y = \log a + b \log x$$

$$\log y = \log a + b \log x$$

$$z = mx + c$$

where  $z = \log y$   $m = b$

$$c = \log a$$

or  $a = e^c$

$$\text{Slope} = b = \frac{z_2 - z_1}{x_2 - x_1} = \frac{\log y_2 - \log y_1}{\log x_2 - \log x_1}$$

### ADVANTAGES OF LINEARIZATION :

Converting non-linear into linear curve has following benefits:  
Linear models are :

- easy to understand
- easy to visualize



- easy to plot
- easy to interpret & analyze
- are more stable and robust
- are easy to compute i.e., by algebraic methods.
- have greater predictability.



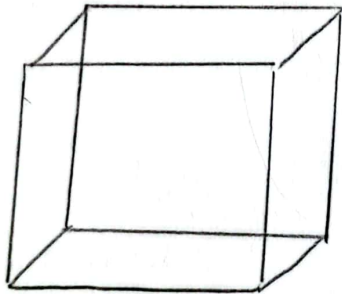
(b)

## • SCALAR TRIPLE PRODUCT:

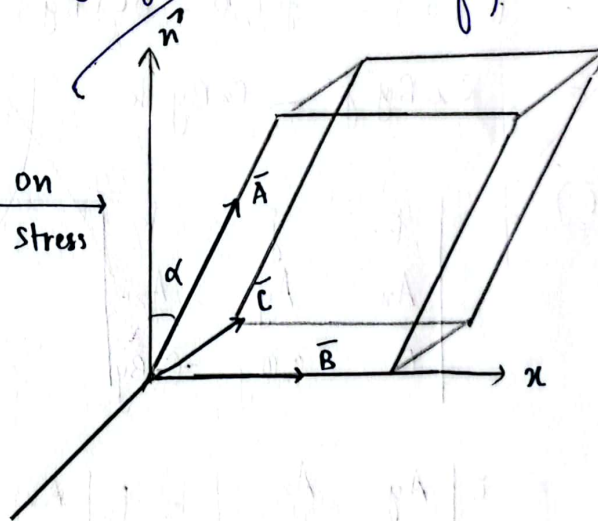
**Definition:** When three vectors are multiplied & their product is scalar, then this multiplication is called STP or scalar triple product.

**Representation:**  $A \cdot (B \times C)$  where  $A, B, C$  are three vectors.

**Physical Meaning:** The physical meaning of STP is area of parallelepiped



on  
stress



**Diagrammatic Representation:**

## • VECTOR TRIPE PRODUCT:

When three vectors are multiplied, and their product is a vector quantity, then such multiplication is called vector triple product.

Consider three vectors  $\vec{A}, \vec{B}, \vec{C}$ , their VTP is rep. HS:  $(\vec{A} \cdot \vec{C})$   
 as  $\vec{A} \times (\vec{B} \times \vec{C})$ , where  $\vec{B} \times \vec{C}$  gives vector,  $\neq \vec{A} \times (\vec{B} \times \vec{C})$  also gives vector

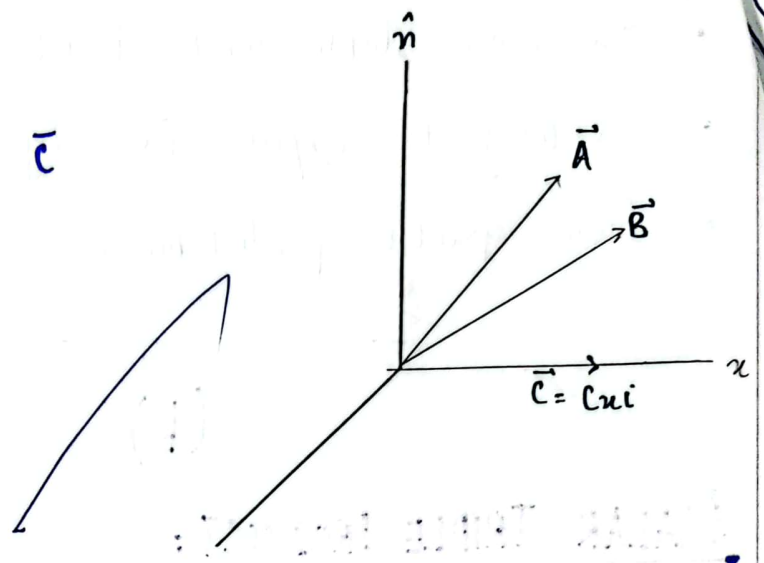
PROOF:

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$\vec{B} = B_x \hat{i} + B_y \hat{j}$$

$$\vec{C} = C_x \hat{i}$$



$$(\vec{B} \times \vec{C}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ B_x & B_y & 0 \\ C_x & 0 & 0 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} B_y & 0 \\ 0 & 0 \end{vmatrix} - \hat{j} \begin{vmatrix} B_x & 0 \\ C_x & 0 \end{vmatrix} + \hat{k} \begin{vmatrix} B_x & B_y \\ C_x & 0 \end{vmatrix}$$

$$= \hat{i} (0) - \hat{j} (0 - 0) + \hat{k} (0 - C_x B_y)$$

$$(\vec{B} \times \vec{C}) = -C_x B_y \hat{k}$$

Now,

$$\vec{A} \times (\vec{B} \times \vec{C}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ 0 & 0 & -C_x B_y \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} A_y & A_z \\ 0 & -C_x B_y \end{vmatrix} - \hat{j} \begin{vmatrix} A_x & A_z \\ 0 & -C_x B_y \end{vmatrix} + \hat{k} \begin{vmatrix} A_x & A_y \\ 0 & 0 \end{vmatrix}$$

$$= \hat{i} (-A_y C_x B_y - 0) - \hat{j} (-A_x C_x B_y - 0) + \hat{k} (0 - 0)$$

$$\vec{A} \times (\vec{B} \times \vec{C}) = -A_y C_x B_y \hat{i} + A_x C_x B_y \hat{j}$$



$\vec{A} \cdot (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$

$$(\vec{A} \cdot \vec{C}) \vec{B} - (\vec{A} \cdot \vec{B}) \vec{C}$$

$$[(A_x \hat{i} + A_y \hat{j} + A_z \hat{k})(C_x \hat{i})](B_x \hat{i} + B_y \hat{j}) - [(A_x \hat{i} + A_y \hat{j} + A_z \hat{k})(B_x \hat{i} + B_y \hat{j})](C_x \hat{i})$$

$$= (A_x C_x)(B_x \hat{i} + B_y \hat{j}) - (A_x B_x + A_y B_y)(C_x \hat{i})$$

$$= A_x C_x B_x \hat{i} + A_x B_y C_x \hat{j} - A_x B_x C_x \hat{i} - A_y B_y C_x \hat{i}$$

$$= -A_y B_y C_x \hat{i} + A_x B_y C_x \hat{j} \quad \text{RHS : LHS}$$

Hence Proved.

3

(c)

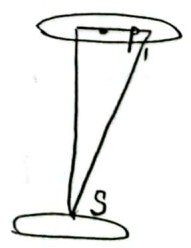
### GRADIENT:

- The multiplication of del operator with scalar quantity ( $\phi$ ), is called gradient.
- It gives shortest distance between two points. i.e. perpendicular.
- It is maximum flow or rate of flow in a direction specified.

### MATHEMATICALLY

$$\Delta = \frac{d}{dx} \hat{i} + \frac{d}{dy} \hat{j} + \frac{d}{dz} \hat{k}$$

$$\Delta(\phi) = \frac{d\phi}{dx} \hat{i} + \frac{d\phi}{dy} \hat{j} + \frac{d\phi}{dz} \hat{k}$$



### DIVERGENCE:

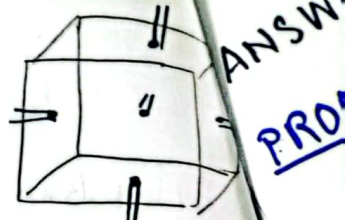
- When  $\Delta$  is multiplied with a vector scalarly, it is known as divergence.
- It is total flux per unit volume.
- It is a vector quantity.

Divergence  $A = \nabla \cdot A$

where  $A$  is a vector,

or  $\boxed{\text{div } A = \nabla \cdot A}$

(dot product)



## CURL:

→ When  $\text{del} (\nabla)$  is multiplied with a vector, vectorically, means its cross product is taken, then it is called curl.

### MATHEMATICALLY

Consider  $\vec{A}$  is a vector,  $A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$

$$\text{curl } \vec{A} = \nabla \times A$$

→ It is total volume per unit area.

## LINE INTEGRAL:

→ The vector  $f(x, y, z)$  along a curve  $AB$ , upon a tangent, is the line integral for  $f(x, y, z)$ .

→ It is the sum of integrals of the component.

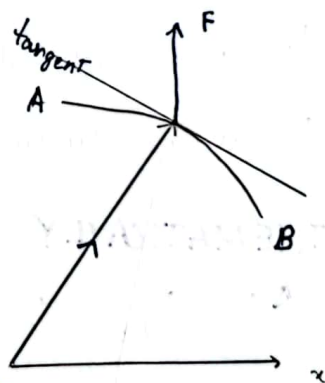
### MATHEMATICALLY:

$$\int L.I = \int \vec{A} \cdot d\vec{r}$$

where  $A$  is force,  $A = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$

$$\text{or } L.I = \int A_x \hat{i} + A_y \hat{j} + A_z \hat{k} \cdot d x \hat{i} + d y \hat{j} + d z \hat{k}$$

$$L.I = \int A_x dx + A_y dy + A_z dz$$



$\hat{i} \cdot \hat{i} = 1$   
 $\hat{j} \cdot \hat{j} = 1$   
 $\hat{k} \cdot \hat{k} = 1$





PROOF:

$$\nabla r^n = n r^{n-2} \vec{r}$$

LHS:

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$r = \sqrt{x^2 + y^2 + z^2} \quad \text{or} \quad r^2 = x^2 + y^2 + z^2$$

$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$r^n = (x^2 + y^2 + z^2)^{n/2}$$

$$\nabla r^n \Rightarrow \frac{d}{dx} \hat{i} + \frac{d}{dy} \hat{j} + \frac{d}{dz} \hat{k} \cdot (x^2 + y^2 + z^2)^{n/2}$$

$$\Rightarrow \frac{d}{dx} (x^2 + y^2 + z^2)^{n/2} \hat{i} + \frac{d}{dy} (x^2 + y^2 + z^2)^{n/2} \hat{j} + \frac{d}{dz} (x^2 + y^2 + z^2)^{n/2} \hat{k}$$

$$\Rightarrow \left[ \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} (2x) \right] \hat{i} + \left[ \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} (2y) \right] \hat{j} + \left[ \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} (2z) \right] \hat{k}$$

$$\Rightarrow nx(x^2 + y^2 + z^2)^{\frac{n-2}{2}} \hat{i} + ny(x^2 + y^2 + z^2)^{\frac{n-2}{2}} \hat{j} + nz(x^2 + y^2 + z^2)^{\frac{n-2}{2}} \hat{k}$$

$$\Rightarrow n \left[ (r^2)^{\frac{n-2}{2}} x \hat{i} + (r^2)^{\frac{n-2}{2}} y \hat{j} + (r^2)^{\frac{n-2}{2}} z \hat{k} \right]$$

$$\Rightarrow n (r^{n-2} x \hat{i} + r^{n-2} y \hat{j} + r^{n-2} z \hat{k})$$

Taking  $r^{n-2}$  common,

$$\Rightarrow n r^{n-2} (x \hat{i} + y \hat{j} + z \hat{k})$$

$$\Rightarrow n r^{n-2} \vec{r} = \text{RHS}$$

Therefore

$$\Delta r^n = n r^{n-2} \vec{r}$$

→ Hence Proved

∵  $x^2 + y^2 + z^2 = r^2$   
Take n common

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

(b)

GIVEN:  $F = 3x^2 \hat{i} + (2xz - y) \hat{j} + z \hat{k}$

$P_1 = (0, 0, 0)$  to  $P_2 = (2, 1, 3)$

$dr = dx \hat{i} + dy \hat{j} + dz \hat{k}$

REQUIRED:  $\int_C \vec{F} \cdot d\vec{r}$

SOLUTION:

$$\begin{aligned} \int \vec{F} \cdot d\vec{r} &= \int [(3x^2 \hat{i} + (2xz - y) \hat{j} + z \hat{k})] dx \hat{i} + dy \hat{j} + dz \hat{k} \\ &= \int 3x^2 dx + (2xz - y) dy + z dz \end{aligned}$$

When 3 points are given, we use 3 point formula:

3 POINT FORMULA

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

$$\frac{x - 0}{2 - 0} = \frac{y - 0}{1 - 0} = \frac{z - 0}{3 - 0}$$

$$\frac{x}{2} = y = \frac{z}{3}$$

$$\frac{x}{2} = t$$

$$y = t$$

$$\frac{z}{3} = t \quad \text{or } z = 3t$$

$$dx = 2 dt$$

$$dy = dt$$

$$dz = 3 dt$$

Lower limit

$$x = 0$$

$$2t = 0$$

$$t = 0$$

Upper limit

$$z = 3$$

$$3t = 3$$

$$t = 1$$

Therefore, on substituting values,

$$\begin{aligned} \int \vec{F} \cdot d\vec{r} &= \int_0^1 [3(2t)^2 (2 dt) + [2(2t)(3t) - t] dt + 3(3t) dt] \\ &= \int_0^1 [6(4t^2) dt + (12t^2 - t) dt + 9t dt] \end{aligned}$$



$$= \int_0^1 24t^2 dt + 12t^2 dt - t dt + 9t dt$$

On integration, we get

$$\therefore \int x^n = \frac{x^{n+1}}{n+1}$$

$$= \left[ \frac{24t^3}{3} + \frac{12t^3}{3} - \frac{t^2}{2} + \frac{9t^2}{2} \right]_0^1$$

$$= \left[ 8t^3 + 4t^3 - \frac{t^2}{2} + \frac{9t^2}{2} \right]_0^1$$

$$= \left[ 12t^3 - \frac{t^2}{2} + \frac{9t^2}{2} \right]_0^1$$

$$= \left[ 12(1)^3 - \frac{(1)^2}{2} + \frac{9(1)^2}{2} \right] - \left[ 12(0)^3 - \frac{0^2}{2} + \frac{9(0)^2}{2} \right]$$

$$= \left( 12 - \frac{1}{2} + \frac{9}{2} \right) - (0 - 0 + 0)$$

$$= 16 \text{ units.}$$

$$\int \vec{F} \cdot d\vec{r} = 16 \text{ Nm}$$

(c)

GIVEN

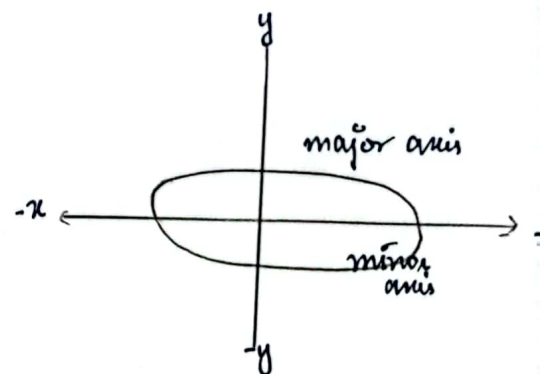
$$A = \pi xy = \text{area of ellipse}$$

$$\% \text{ of } x = 0.01 \text{ of } x$$

$$\% \Delta x = 0.01 x$$

$$\% \text{ of } y = 0.01 \text{ of } y$$

$$\% \Delta y = 0.01 y$$



REQUIRED:

$$\% \text{ error in } A = ?$$

### SOLUTION :

$$A = \pi xy$$

$$\frac{\Delta A}{A} = \frac{\Delta x}{x} + \frac{\Delta y}{y}$$

$$\frac{\Delta A}{A} = \frac{0.01x}{x} + \frac{0.01y}{y}$$

$$\Delta A/A = 0.01 + 0.01$$

$$\Delta A/A = 0.02$$

For % error :

$$\frac{\Delta A}{A} \times 100 = 0.02 \times 100$$

$$\% \text{ error in } A = 2\%$$

### RESULT :

Therefore, % error in area of ellipse = 2%

3

The End



(SE-080)

# NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF SOFTWARE ENGINEERING

FIRST YEAR - BATCH 2024

MID TERM EXAMINATIONS – SPRING 2024

Duration: 90 Minutes

Date: 22-05-2024

Max. Marks: 20

## ES-205 (Islamic Studies)

Note: Attempt all of the given questions. Read the questions carefully and answer accordingly.

**Q1.** Describe essential details about the Event of Ascension. Also, share your understanding of the event of Ascension. And make a biographical chart from 4<sup>th</sup> year of prophethood to 13<sup>th</sup> year of prophethood. 9

CLO# 2 (10 Marks)

**Q2.** Translate all of the sacred texts and discuss the concept of Tauheed and different aspects in the light of the given texts of the holy Quran. 9.5

CLO# 1 (10 Marks)

1. لَوْ كَانَ فِيهِمَا آلِهَةٌ إِلَّا اللَّهُ لَفَسَدَتَا فَسُبْحَانَ اللَّهِ رَبِّ الْعَرْشِ عَمَّا يَصِفُونَ ﴿٢٢﴾ (الأنبياء: 22)
2. وَالْهَكْمُ إِلَهٌ وَاحِدٌ لَا إِلَهَ إِلَّا هُوَ الرَّحْمَنُ الرَّحِيمُ ﴿١٦٣﴾ (البقرة: 163)
3. إِنَّ فِي خَلْقِ السَّمَاوَاتِ وَالْأَرْضِ وَاخْتِلَافِ اللَّيْلِ وَالنَّهَارِ وَالْفُلْكِ الَّتِي تَجْرِي فِي الْبَحْرِ بِمَا يَنْفَعُ النَّاسَ وَمَا أَنْزَلَ اللَّهُ مِنَ السَّمَاءِ مِنْ مَاءٍ فَأَخْيَا بِهِ الْأَرْضَ بَعْدَ مَوْتِهَا وَبَثَّ فِيهَا مِنْ كُلِّ دَابَّةٍ وَتَصْرِيفِ الرِّيَّاحِ وَالسَّحَابِ الْمُسَخَّرِ بَيْنَ السَّمَاءِ وَالْأَرْضِ لَآيَاتٍ لِقَوْمٍ يَعْقِلُونَ ﴿١٦٤﴾ (البقرة: 164)

ANSWER NO. 1

## EVENT OF ASCENSION :

The miraculous night journey in which Prophet travelled from Al-Aqsa (in Jerusalem) to heavens is known as 'Al-Miraj' or 'Ascension'.

### MENTION:

It is mentioned in Surah Al-Najm.

COMPANY: Hazrat Jibrael (عليه السلام)

MEANS: Al-Buraaq (a horse - miraculous)

### ESSENTIAL DETAILS:

- ➔ FIRST HEAVEN: The angel of heaven opened the gate & welcome him, there He met Hazrat Adam (عليه السلام), and martyrs were on the right, while wretched were on the left.
- ➔ 2<sup>ND</sup> HEAVEN: Prophet (PBUH) met John, (Son of Zakariyyah), or also known as Hazrat Yahya, and son of Mary, Jesus. Both paid tributes to Prophet (ﷺ).
- ➔ 3<sup>RD</sup> HEAVEN: There, he met Joseph, Son of Yaqub (Yusuf (عليه السلام)). He saluted our beloved Prophet.
- ➔ 4<sup>TH</sup> HEAVEN: Prophet (PBUH) met Hazrat Idris (Enoch) who paid his regards, which were reciprocated by Hazrat Muhammad (ﷺ).
- ➔ 5<sup>TH</sup> HEAVEN: He met Hazrat Aaron (Haaron), brother of Moses who saluted him.
- ➔ 6<sup>TH</sup> HEAVEN: Hazrat Muhammad (SAWW) met Moses at the 6<sup>th</sup> heaven. Moses wept by seeing the person who has instructed much more people than him and who has come in to this world after his departure.



→ 7th HEAVEN: Prophet met Hazrat Ibrahim (عليه السلام) who is the father of Prophets, saluted him and moved further.

→ After this, he reached Sidra-tul-Muntaha (the remotest lote tree) and also visited Bait-ul-Mamoor, where Angels worship. Each day, more than 75,000 angels come to worship Allah, and their turn will not come again till Judgement Day.

→ Afterwards, Holy Prophet (PBUH) experienced Divine Mercy. The presence of Divine Glory. On the return, he told Moses about the 50 prayers that is a reward upon his Ummah. Moses said: "Your people won't be able to offer, lessen these prayers." Prophet PBUH after consulting with Hazrat Jibreel went & asked for reduction. Then told Moses, that now the number is ten. He again said to go back. At last, the prayers became 5. Moses still asked for reduction but Prophet said, "I am ashamed now."

#### OTHER IMPORTANT EVENTS:

1. Prophet visited two rivers, Nile and Euphrates and two hidden rivers.
2. Prophet PBUH was presented with milk and wine. Out of which he selected milk. He attained Al-Fitrah.
3. He met angel of Hell, Malik, & saw Hell dwellers

you know?

a

# (MAKKAN PHASE) PROPHETHOOD (4-13 Years)

4<sup>th</sup> - 10<sup>th</sup> Years

Open Preaching within Makkah

- 1<sup>st</sup> Migration to Abyssinia  
(12 men, 4 women)
- 2<sup>nd</sup> Migration to Abyssinia  
(83 men, 18 or 19 men)
- Revelation of Verse 214, of Chap Ashura (The Poets)  
'O Prophet, warn tribe of near kindred' (26:214)
- Called Bani Fahr' and Bani Adi in Mount As-Safa.
- Revelation of Verses of Surah Lahab
- Polytheists reaction (15:94)
- Al-Waleed bin Al-Mugheerah's Confederacy.  
(Poet - Kahin - Majnun - witch)  
(V 74: 18-25)
- Barter of Ammar bin Al-Walid bin Al-Mugheerah  
(Rejected by Abu Talib)
- Conversion of Hamza bin Abd-al-Muttalib & Umar bin Ab-Khattab  
(6<sup>th</sup> Dhul-hijjah) (10<sup>th</sup> Year)
- FIRST AQABAH PLEDGE  
(Sin Madanese from Zhanzraj tribe)
- Shib-e-Talib (3 Years) (Siege)  
(7<sup>th</sup> to 10 Years)
- Year of Grief (Deaths of Abu Talib (Rafab) Khadija (Rarnadan)) ] 10<sup>th</sup> Year

11<sup>th</sup> - 13<sup>th</sup> Years

Open Preaching beyond Makkah

- Journey to At-Taif  
(50 km away - Companion: Zaid bin Haritha - 10 days stay)  
(Shawwal, 620 AD, exact 10 years after revelation)
- Stay in Waadi Naklah  
(Jins listened to Quran)  
(46: 29-31)
- Meeting with Jibrael and Angel of mountains
- Second Aqabah Pledge  
(12 men)
- Third Aqabah Pledge (13<sup>th</sup> yr)  
(73 men, 2 women  
• Nuseibah from Najjar  
• Asma from Bani Salamah)
- Isra-al-Miraj Journey  
(10-11<sup>th</sup> year)
- Dar-al-Nadwa  
(Assassination Plan by Jehal)



1. **VERSE 22** (Chapter Al-Ambüya)

Translation:

"If there were gods besides Allah, there would have been confusion in both: So glory be to Allah, the Lord of Throne from what they attribute."

Subject Matter: 'Unity of God'

Interpretation:

1. If there were two or more than two Gods, there would be never the Unity of Thought among them.
2. A God with divided sovereignty is not a God.
3. Conflicts will result in chaos and destruction of system of land & heaven

CONCLUSION { This verse nullify the concept the multiple Gods and promotes monotheism, and Tauheed. As, the proper functioning of system ensures that there is Supreme Authority who is managing affairs of all sorts.

2. **VERSE 163** (Chapter Al-Baqara)

Translation:

"And your Lord is One Lord, there is no God except He, the Most-Beneficent and the Most-Merciful." (2:163)

Subject: "Tauheed &amp; Shirk"

Different Aspects / Interpretations:

The verse consists of three segments, which are defined as follows.

1<sup>st</sup> SEGMENT: ONENESS

Surah Ikhlaas gives the better and best understanding of this segment. It is also known as Surah Tauheed.

"Say, He is Allah - The One. Allah is Besought of All, needs no one. He neither begets nor was begotten. There is none comparable to Him."

## 2<sup>nd</sup> Segment: Denial of Polytheism

Polytheism has further three types or domains, and are as follows:

- WORDSHIP: There is only One God, claiming multiple Gods is the form of Shirk, which is a major sin.
- ATTRIBUTES: Assigning Divine Attributes to any other deity, except Allah is known as polytheism in attributes.
- WORSHIP: Loving and compelling to someone else, other than Allah is Polytheism in worship. If the intentions are not for Allah, and are for showcasing, and other purpose, also includes in the Shirk.

## 3<sup>rd</sup> Segment: ATTRIBUTES:

- الرَّحْمَنُ : Being Merciful towards All humans
- الرَّحِيمُ : Being Merciful towards Believers

## CONCLUSION:

One must not associate any Partner with Allah, as Shirk is 'Zulmun - Adheem', as declared in Quran, by Allah, Himself. His Mercy, should not make us do sins more, but should increase our reverence, love, and respect from Him.

*"And My Mercy encompasses all things" (Al-Quran)*

## 3. VERSE 164 ( CHAPTER AL BAQARA )

Translation:

*"Surely, in the creation of heavens and earth, and in the alternation of night and day, and in the sail of ships in sea, carrying that which benefits men, and the rain that is sent down from sky, thereby reviving earth, after which it was dead; and in the spreading of creatures and the winds that blows, and all that which is manifested between the heaven and earth; there are signs for those who have sense." (2: 164)*



## Subject: "TANGIBLE EXAMPLES OF TAUHEED"

### Interpretation:

People who contemplate over surroundings, who have sense, can clearly see the signs of Allah. There are enough reasons to believe that there is an authority who is the Manager of whole universe. People having sense of knowledge can easily understand these verses.

### CONCEPT OF TAUHEED, IN ENLIGHTENMENT OF VERSES:

- ① Allah is One in His State of being.
- ② He is the anterior and posterior. (He existed when nothing did & will remain, when nothing would)
- ③ He is free from conceivable things (not made or consist of atoms, plasma, particles).
- ④ He is the One claiming all rights of worship.
- ⑤ He is Self-sufficient (غني) (He doesnot need our worship, but we need it).

Neat work

a.s

The End